Wild-place metaplectic character transfer in degree two: Toral positivity and the nonabelian gluing problem

Draft 0 — standalone companion to Part IV

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Abstract

We initiate a separate local project on irreducible-character integrality for a scalar-central odd-degree metaplectic cover of the quaternion division algebra. The target is the still-open wild-place assertion that the unitary division-side transform of Part III carries irreducible characters to genuine irreducible characters. We distinguish this conjecture from a proved uniform quadratic toral permutation-character theorem: for every quadratic E/F in odd residue characteristic, and every odd n with $\mu_n \subset F$, the discriminant ratio $\Delta(t)/\Delta(t^n)$ is a finite permutation character of $E^\times/F^\times$. A complete worked example over $F = \mathbb{Q}_3(\zeta_3)$ gives positive genuine torus multiplicities $(3, 2, 2, 2)$ in the unramified case and $(3, 2, 1, 1, 1, 1)$ for either ramified quadratic torus, each of total degree 27. We prove an integrality–positivity criterion for the entire unitary character matrix, explain why toral positivity does not prove global nonabelian positivity, and give a staged program using the explicit cocycle, finite twisted quotients, Clifford theory, and exact character tests. No genuine 27-dimensional irreducible of the full covered division algebra is constructed in this draft.

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1 Purpose, scope, and status

The wild division-side character problem is separated from the conditional global trace comparison in Part IV of the larger series. The following results are the starting point, with proofs either included here or explicitly attributed to earlier papers. The remaining full-group integrality assertion is a *conjecture*, not a theorem.

Statement	Source	Status
Unitary local L_D and identity factor	Part III, Draft 9	[E]
Heisenberg commutator and degree on maximal tori	Part I, Draft 8	[E]
Quadratic toral permutation-character theorem	This paper, Theorem 4.1	[P]
Three exact cubic wild field-torus calculations	Section 6	[P]
All-row positivity–integrality equivalence	Proposition 3.1	[P]
Irreducible-character integrality on $\widetilde{D^\times}$	Section 7	[C]

The running example

The local field $F = \mathbb{Q}_3(\zeta_3)$ studied here is the wild place $\mathfrak{p}_3 = (1-\zeta_3)$ of the series' running example $K = \mathbb{Q}(\zeta_3)$, $n = 3$, $r = 2$, $D = (-1, 4-\zeta_3)_K$, which is ramified exactly at $\mathfrak{p}_3$ and $\mathfrak{p}_7 = (3+\zeta_3)$ (see `notes/mjl_running_example.tex` in the project repository). Since $3+\zeta_3 \equiv 1 \pmod{\mathfrak{p}_3}$ is a square, $D_{\mathfrak{p}_3} \cong (-1, \varpi)$, which is exactly the model $E_0 \oplus E_0 j$ with $j^2 = \varpi$ of Example 6.1. Target A of §7 is therefore the first open case of wild integrality in the running example.

2 Division-side character transform and its normalization

Let F be nonarchimedean, D/F a division algebra of prime degree r, $H = D^\times$, and $\widetilde{H}$ its specified cover. Set $Z = F^\times$ and $\widetilde{Z} = p^{-1}(Z)$, equipped with compatible Haar measures on $H/Z \simeq \widetilde{H}/\widetilde{Z}$. The Part III transform is

$$(L_D\Phi)(z\,t^*) = b_D\,\widetilde{\theta}(z)\frac{\Delta(t)}{\Delta(t^n)}\Phi(t), \qquad b_D = |n|_F^{-(r-1)/2}, \tag{1}$$

for regular t, $z \in \widetilde{Z}$; it vanishes at nongood regular lifts. Here

$$\Delta(t) = \left|\det(1-\mathrm{Ad}(t) \mid \mathfrak{h}/\mathfrak{t})\right|_F^{1/2} = \prod_{i<j}\left|\frac{(\lambda_i-\lambda_j)^2}{\lambda_i\lambda_j}\right|_F^{1/2}. \tag{2}$$

This equals Part I's $|D(t)|^{1/2}|\mathrm{Nrd}(t)|_F^{-(r-1)/2}$. Since the power map on a field torus modulo scalars has index $|n|_F^{-(r-1)}$, Weyl integration gives b_D^2 equal to this index. The factor b_D is independent of

common rescalings of the two identified quotient Haar measures. The ratio of discriminants extends at central elements to $|n|_F^{-r(r-1)/2}$ (Part III, Lemma 3.3), hence

$$(L_D\Theta_\rho)(1) = c_D \dim \rho, \qquad c_D = b_D|n|_F^{-r(r-1)/2} = |n|_F^{-(r^2-1)/2}. \tag{3}$$

The identity factor is pointwise and does not depend on Haar measures. Part III, Theorem 9.1, separately gives the adjoint orbital-integral/test-function transfer. Test functions transform by $\widetilde{\theta}^{-1}$, not by $\widetilde{\theta}$.

2.1 Integral-matrix criterion

Choose $\widetilde{\theta}$ first and put $\theta(a) = \widetilde{\theta}(a^*)$. Part III, Theorem 4.3 and Corollary 5.1, show that L_D is a unitary *isomorphism* of invariant L^2 spaces and that

$$L_D\Theta_\rho = \sum_\tau a_{\rho\tau}\Theta_\tau, \qquad a_{\rho\tau} = \langle L_D\Theta_\rho, \Theta_\tau\rangle, \tag{4}$$

with each row and column finite. The characters on either side are orthonormal bases in their fixed-central-character spaces.

Proposition 2.1 ([**E**] Integral-lattice criterion, Part III)**.** *The transform L_D permutes the irreducible characters with positive signs if and only if every $a_{\rho\tau}$ in (4) is an integer.*

Proof. An integral unit vector in ℓ^2 is a signed basis vector; unitarity gives injectivity and surjectivity because the inverse is the adjoint of a unitary map. The value at identity in (3) is positive and fixes the sign. This is Part III, Corollary 5.2, with its converse. □

Remark 2.2 (The actual obstruction)**.** For wild v, Part III does not prove $a_{\rho\tau} \in \mathbb{Z}$. The tame Adams argument works with a finite quotient Q and a central subgroup C, under the hypothesis $\gcd(n, |Q/C|) = 1$ (Part III, Lemma 7.5). At a wild place that hypothesis need not hold. For the simplest illustration take $Q = C_p$, $C = 1$, $n = p$: then the Adams operation sends every linear character to the trivial character. This does not prove a nonintegral coefficient for L_D; it only shows why that particular proof method fails.

3 A useful global criterion: positivity of every row

Proposition 3.1 ([**P**] Positivity versus integrality)**.** *With the unitary character matrix $(a_{\rho\tau})$ of (4), the following are equivalent:*

(a) L_D permutes irreducible characters with positive signs;

(b) every $a_{\rho\tau}$ is an integer;

(c) every $a_{\rho\tau}$ is a nonnegative real number;

(d) every $L_D\Theta_\rho$ is positive definite on $\widetilde{H}$.

The implication from positive definiteness to integrality requires every *row; positivity of a single row does not force integrality or irreducibility for that row.*

Proof. By Part III, each row and column of A has ℓ^2 norm one. An integer row is a signed basis vector, and its sign is positive by the identity value (3). If all entries are nonnegative, orthogonal rows have disjoint supports. Every column meets exactly one row, and since its norm is one its nonzero entry is one; row norms then imply the permutation property. The equivalence of nonnegative character coefficients and positive definiteness follows from Peter–Weyl/GNS for a compact-mod-center group with fixed unitary central character. One row could instead be $(2^{-1/2}, 2^{-1/2}, 0, \ldots)$. □

Remark 3.2. At a finite twisted quotient the GNS condition can be checked by exact positivity of the kernel $(\phi(x^{-1}y))_{x,y}$. A positive-semidefinite test for one ρ verifies positivity, *not* the integral-character claim for that ρ. Computing the actual coefficients or obtaining a single-row virtual-character theorem remains indispensable unless positivity is proved uniformly for all rows.

4 A general quadratic theorem

Let F be a finite extension of $\mathbb{Q}_p$ with p odd, residue field of cardinality q, and let $n \geq 1$ be odd with $\mu_n \subset F$. Let E/F be any quadratic field extension, with nontrivial Galois involution σ. Put

$$A = E^\times/F^\times, \qquad G = E^1 = \ker(N_{E/F} : E^\times \to F^\times).$$

Hilbert's Theorem 90 identifies $A \xrightarrow{\sim} G$ via $t \mapsto u = t/\sigma(t)$. The normalized absolute value on E is $|x|_E = q_E^{-v_E(x)}$, so $|x|_E = |N_{E/F}x|_F$.

For noncentral $t \in E^\times$ let

$$R_E(t) = \frac{\Delta(t)}{\Delta(t^n)}, \qquad \Delta(t) = \left|\det(1 - \mathrm{Ad}(t) \mid D/E)\right|_F^{1/2}.$$

The ratio extends continuously at scalar elements, with value $|n|_F^{-1}$.

Theorem 4.1 (**[P]** Quadratic toral permutation positivity)**.** *For any $E/F, n$ as above, R_E is the character of a finite-dimensional permutation representation of a finite quotient of A. In particular,*

$$R_E(t) = \sum_{\chi \in \widehat{A}} m_\chi \chi(t), \qquad m_\chi \in \mathbb{Z}_{\geq 0},$$

where only finitely many m_χ are nonzero. This includes every wild case $p \mid n$. At the identity $R_E(1) = |n|_F^{-1}$.

4.1 A radial valuation formula

Write $n = p^a m$ with $(m, p) = 1$, and put

$$t_\zeta = v_E(1 - \zeta), \qquad G_k = \{u \in G : v_E(1 - u) \geq k\} \quad (k \geq 1), \qquad G_0 = G.$$

If $a = 0$ then $R_E \equiv 1$, so assume $a \geq 1$.

Lemma 4.2. *For $u \in G$ and any $\zeta \in \mu_n$ whose prime-to-p component is nontrivial, $|1 - \zeta u|_E = 1$. For $\zeta \in \mu_{p^a} \setminus \{1\}$,*

$$v_E(1 - \zeta u) = \min\{v_E(1 - u), t_\zeta\}.$$

Consequently, with $k = v_E(1 - u) \in \mathbb{Z}_{\geq 0} \cup \{\infty\}$,

$$R_E(u) = q_E^{\frac{1}{2}\sum_{\zeta \in \mu_{p^a} \setminus \{1\}} \min(k, t_\zeta)}. \tag{5}$$

Proof. The reduction of $u \in E^1$ lies in a norm-one subgroup. In the unramified case its intersection with $k_F^\times$ is $\{\pm 1\}$, and in the ramified case the reduction of u is itself in $\{\pm 1\}$. A root of unity with nontrivial prime-to-p component has reduction in $k_F^\times \setminus \{\pm 1\}$, since n is odd, proving the first assertion.

For a nontrivial p-power root ζ, the ultrametric inequality gives the asserted minimum when the two valuations differ. If both equal t_ζ and $v_E(1-\zeta u) > t_\zeta$, then $u \equiv \zeta^{-1} \pmod{\mathfrak{p}_E^{t_\zeta+1}}$. Applying σ and using $\sigma(u) = u^{-1}$ gives $u^{-1} \equiv \zeta^{-1}$ to the same precision, hence $\zeta \equiv \zeta^{-1}$, a contradiction to $v_E(\zeta - \zeta^{-1}) = t_\zeta$ (as p is odd). Thus equality holds.

For $u = t/\sigma(t)$ we have $\Delta(t) = |N_{E/F}(1-u)|_F^{1/2} = |1-u|_E^{1/2}$. Factor $1-u^n = \prod_{\zeta\in\mu_n}(1-\zeta u)$ and use the two valuation statements. The identity follows for noncentral t and extends continuously to $u = 1$. At $u = 1$, the cyclotomic product $\prod_{\zeta\in\mu_{p^a}\setminus\{1\}}(1-\zeta) = p^a$ gives $R_E(1) = |p^a|_E^{-1/2} = |n|_F^{-1}$. $\square$

Define for $k \geq 0$

$$M_k = q_E^{\frac{1}{2}\sum_{\zeta\in\mu_{p^a}\setminus\{1\}} \min(k,t_\zeta)}, \qquad M_0 = 1,$$

and put $T = \max_{\zeta\neq 1} t_\zeta$. All values M_k are positive integers because roots ζ, ζ^{-1} occur in pairs of equal valuation. We have $M_k = M_T$ for $k \geq T$.

4.2 The unramified filtration

Suppose E/F is unramified. Then $q_E = q^2$ and reduction of the norm-one group has order $q+1$; at every deeper level the trace-zero subgroup of the residue-field additive group has order q. Thus the filtration satisfies

$$[G : G_1] = q+1, \qquad [G : G_k] = (q+1)q^{k-1} \quad (k \geq 1).$$

Write $s_k = \#\{\zeta \in \mu_{p^a} \setminus \{1\} : t_\zeta \geq k\}$. Each s_k is even, and $M_k/M_{k-1} = q^{s_k}$. Set

$$c_k = \frac{M_k - M_{k-1}}{(q+1)q^{k-1}} \quad (1 \leq k \leq T).$$

For $k = 1$, $M_0 = 1$ and $q+1$ divides $q^{s_1} - 1$ because s_1 is even. For $k > 1$ with $s_k > 0$, M_{k-1} is divisible by q^{k-1}, since at least two roots contribute at least $k-1$ to its exponent. Again $q+1$ divides $q^{s_k} - 1$. Hence $c_k \in \mathbb{Z}_{\geq 0}$ (and $c_k = 0$ if $s_k = 0$).

By telescoping (5),

$$\boxed{R_E = \mathbf{1}_G + \sum_{k=1}^{T} c_k\,[G : G_k]\,\mathbf{1}_{G_k}.} \tag{6}$$

Here $[G : G_k]\,\mathbf{1}_{G_k}$ is precisely the permutation character of G on G/G_k.

4.3 The ramified filtration

Suppose E/F is ramified. Then $q_E = q$ and $t_\zeta = 2v_F(1-\zeta)$ is even. The norm-one group has filtration

$$[G : G_1] = 2, \qquad G_{2j} = G_{2j+1}, \qquad [G : G_{2j}] = 2q^j \quad (j \geq 1).$$

For completeness, write $E = F(\sqrt{d})$ with $v_F(d) = 1$. Hilbert 90 gives $u = (a + b\sqrt{d})/(a - b\sqrt{d})$; normalize a, b to be integral and not both divisible by the uniformizer of F. If a is a unit, scale it to 1 and write $c = b/a \in \mathcal{O}_F$: then the depth of $u - 1$ is $1 + 2v_F(c)$. Moreover, two such parameters c, c' define elements congruent modulo $\mathfrak{p}_E^{2j}$ precisely when $c \equiv c' \pmod{\mathfrak{p}_F^j}$, so $[G_1 : G_{2j}] = q^j$. If a

is not a unit and b is, then $u \equiv -1$ modulo $\mathfrak{p}_E$. These observations give the displayed indices and the fact that positive depths are odd.

Define

$$c_1 = \frac{M_1 - 1}{2}, \qquad c_{2j} = \frac{M_{2j+1} - M_{2j-1}}{2q^j} \quad (1 \le j \le T/2).$$

The first coefficient is an integer because $M_1 = q^{(p^a-1)/2}$ is odd. For $j \ge 1$ write

$$\frac{M_{2j+1}}{M_{2j-1}} = q^{e_j}, \qquad e_j = \frac{1}{2}\sum_{\zeta \ne 1} (\min(2j+1, t_\zeta) - \min(2j-1, t_\zeta)) \in \mathbb{Z}_{\ge 0}.$$

If $e_j = 0$, then $c_{2j} = 0$. Otherwise some pair $\{\zeta, \zeta^{-1}\}$ has $t_\zeta \ge 2j$, so M_{2j-1} is divisible by q^{2j-1}, hence by q^j. Moreover $q^{e_j} - 1$ is even. Therefore $c_{2j} \in \mathbb{Z}_{\ge 0}$.

As all positive depths are odd, telescoping now gives

$$\boxed{R_E = \mathbf{1}_G + c_1\,[G : G_1]\mathbf{1}_{G_1} + \sum_{j=1}^{T/2} c_{2j}\,[G : G_{2j}]\,\mathbf{1}_{G_{2j}}.} \tag{7}$$

Again each summand is the character of a permutation representation. Equations (6) and (7) prove Theorem 4.1.

5 From Fourier coefficients to covered-torus restrictions

Lemma 5.1 ([**P**] Fourier test for restrictions to covered field tori)**.** *Let E/F be any maximal field in D, with r prime and $(r, n) = 1$, and put $A_E = E^\times/F^\times$. The function*

$$R_E(t) = \frac{\Delta(t)}{\Delta(t^n)}$$

extends continuously at scalars, is a locally constant function on the compact abelian group A_E, and has a finite Fourier expansion

$$R_E(t) = \sum_{\chi \in \widehat{A_E}} m_\chi \chi(t), \qquad m_\chi = \int_{A_E} R_E(t)\overline{\chi(t)}\,dt. \tag{8}$$

Here the Haar measure of A_E has mass 1. Choose a compatible genuine central character $\widetilde{\theta}$ of $\widetilde{Z}$ and set $\theta(a) = \widetilde{\theta}(a^*)$. For any smooth character $\alpha : E^\times \to \mathbb{C}^\times$ with $\alpha|_{F^\times} = \theta$, and for each $\chi \in \widehat{A_E}$, let $\eta_{\alpha\chi}$ be the unique genuine irreducible representation of $\widetilde{E^\times}$ whose central character satisfies

$$\zeta_{\alpha\chi}(z\,t^*) = \widetilde{\theta}(z)\alpha(t)\chi(t) \qquad (z \in \widetilde{Z},\ t \in E^\times).$$

Define a torus-level class function $\mathcal{L}_E\alpha$ to equal $b_D\widetilde{\theta}(z)R_E(t)\alpha(t)$ on $zt^* \in C_E$ and to vanish outside C_E. Then, as a genuine class function on the covered torus*,*

$$\mathcal{L}_E\alpha = \sum_\chi m_\chi\,\Theta_{\eta_{\alpha\chi}}. \tag{9}$$

For an ordinary character Θ_ρ of $D^\times$, its actual L_D transform restricts to the sum of these $\mathcal{L}_E\alpha$ with the multiplicities in $\Theta_\rho|_{E^\times}$. Consequently $m_\chi \in \mathbb{Z}_{\ge 0}$ for every χ is sufficient for the restriction of every transferred ordinary irreducible character to this covered torus to be a genuine representation character. It is not sufficient for the full $\widetilde{D^\times}$ character to be a representation character.

Proof. The extension of R_E at scalars follows from Part III, Lemma 3.3. Since E/F is a degree-r field extension with $(r,n)=1$, the nth power of a noncentral t stays noncentral. The compactness of A_E and local constancy imply that R_E factors through a finite quotient, giving (8).

By Part I the center of the covered torus is $C_E = \widetilde{Z} \cdot \{t^* : t \in E^\times\}$. The displayed formula for $\zeta_{\alpha\chi}$ is well-defined: the power map on A_E is injective, and on scalar overlaps the compatibility $\alpha(a) = \widetilde{\theta}(a^*)$ is precisely what is required. Such genuine central characters extend over C_E, and Stone–von Neumann gives a unique $\eta_{\alpha\chi}$. Its character vanishes off C_E, while on C_E it equals $b_D\zeta_{\alpha\chi}$ because its degree is b_D. The defining formula (1) for L_D has the identical vanishing and equals $b_D\widetilde{\theta}(z)R_E(t)\alpha(t)$ on zt^*. Comparing with (8) proves (9). Finally, restricting a finite-dimensional ordinary representation to $E^\times$ decomposes it into one-dimensional characters, all restricting to θ on $F^\times$; summing the displayed formula proves the last claim. □

6 The explicit cubic wild example

The following calculation is more informative than the dimension obstruction alone. It verifies a *necessary representation-ring integrality test on every maximal torus*, including the two ramified quadratic field types; the full-group integrality statement remains open.

Example 6.1 (The field, algebra, and cover)**.** Take

$$F = \mathbb{Q}_3(\zeta_3), \qquad \varpi = \zeta_3 - 1, \qquad 3 = -\zeta_3^{-1}\varpi^2, \qquad q_F = 3.$$

Thus $v_F(3) = 2$ and $|3|_F = 3^{-2} = 1/9$. The compatible covering parameters in [JC] and Part I are respectively $c = 1$ and $c_v = 2 \pmod 3$. Let D/F be the unique quaternion division algebra. With $E_0 = F(i)$, $i^2 = -1$, its unramified quadratic maximal subfield, we may write

$$D = E_0 \oplus E_0 j, \qquad j^2 = \varpi, \qquad jx = \sigma(x)j \quad (x \in E_0),$$

where σ is the nontrivial automorphism of E_0/F. Part I, Proposition 3.5, gives, for every quadratic maximal subfield E/F, the covered-torus commutator

$$[x,y]_{\widetilde{E^\times}} = (N_{E/F}x, N_{E/F}y)^2_{F,3}(x,y)^{-1}_{E,3}. \tag{10}$$

The center C_E of $\widetilde{E^\times}$ is the preimage of $F^\times E^{\times 3}$, and

$$[E^\times : F^\times E^{\times 3}] = 9, \qquad b_D = 3, \qquad c_D = 27.$$

By the Heisenberg-type Stone–von Neumann theorem ([JC], Proposition 2.4; Part I, Proposition 3.5), every genuine irreducible representation of $\widetilde{E^\times}$ has degree 3. This is *not* a degree calculation for representations of $\widetilde{D^\times}$.

6.1 The unramified quadratic torus

Put $E_0 = F(i)$ as above. Since X^2+1 is irreducible over $\mathbb{F}_3$, E_0/F is unramified and its residue field is $\mathbb{F}_9$. For $t = a + bi$ with $a, b \in F$ the reduced norm and trace give the elementary identity

$$R_{E_0}(a+bi) = \frac{|a^2+b^2|_F}{|3a^2-b^2|_F}. \tag{11}$$

Multiplying t by an element of $F^\times$, suppose $a, b \in \mathcal{O}_F$ and at least one is a unit. Then $a^2 + b^2$ is always a unit. If b is a unit, so is $3a^2 - b^2$. If $b = \varpi u$, then

$$3a^2 - b^2 = 3(a^2 + \zeta_3 u^2),$$

and a is a unit; the reduction of $a^2 + \zeta_3 u^2$ modulo ϖ is either 1 or 2, never 0. Hence

$$R_{E_0}(a + bi) = \begin{cases} 9, & b \in \varpi\mathcal{O}_F, \\ 1, & b \in \mathcal{O}_F^\times. \end{cases} \tag{12}$$

The finite residue quotient is

$$B_0 = k_{E_0}^\times / k_F^\times \simeq C_4.$$

The condition $b \in \varpi\mathcal{O}_F$ says exactly that the image of t in B_0 is the identity. Denote its four characters by $\chi_0 = 1, \chi_1, \chi_2, \chi_3$. The regular character of C_4 equals 4 at the identity and 0 elsewhere. Thus

$$\boxed{R_{E_0} = 1 + 2\,\mathrm{reg}_{C_4} = 3\chi_0 + 2\chi_1 + 2\chi_2 + 2\chi_3.} \tag{13}$$

This is a positive integral Fourier decomposition of degree $3 + 2 + 2 + 2 = 9$; it is stronger than merely checking that R_{E_0} takes integer values.

6.2 Both ramified quadratic tori

Every ramified quadratic extension of F can be written $E_d = F(j_d)$, $j_d^2 = d = u\varpi$ with $u \in \mathcal{O}_F^\times$. There are two isomorphism classes, distinguished by the square class of u. For $t = a + bj_d$,

$$R_{E_d}(a + bj_d) = \frac{|a^2 - db^2|_F}{|3a^2 + db^2|_F}. \tag{14}$$

Again normalize a, b to be integral and not both divisible by ϖ. The terms $3a^2$ and db^2 have different valuations whenever both a and b are units. A direct valuation calculation therefore gives

$$R_{E_d}(a + bj_d) = \begin{cases} 9, & a \in \mathcal{O}_F^\times, \quad b \in \varpi\mathcal{O}_F, \\ 3, & a, b \in \mathcal{O}_F^\times, \\ 1, & a \in \varpi\mathcal{O}_F, \quad b \in \mathcal{O}_F^\times. \end{cases} \tag{15}$$

For example, in the original cyclic algebra $d = \varpi$,

t	j	$1 + j$	$1 + \varpi j$
$R(t)$	1	3	9
$(L_D 1)(t^*)$	3	9	27

for the central-character normalization of Example 6.1. The three values illustrate why the unramified torus alone is not an adequate test.

There is a natural finite quotient

$$B_d = E_d^\times / (F^\times U_1) \simeq C_2 \times C_3,$$

where $U_1 \subset \mathcal{O}_{E_d}^\times$ consists of $a + bj_d$ with $a \in \mathcal{O}_F^\times$ and $b \in \varpi\mathcal{O}_F$; equivalently $U_1 = \mathcal{O}_F^\times(1 + \mathfrak{p}_{E_d}^2)$. The C_2 coordinate records the parity of the E_d-valuation, and on even elements the C_3 coordinate is $\bar{c} \in \mathbb{F}_3$ for $1 + cj_d$ with $c \in \mathcal{O}_F$ read modulo ϖ. The multiplication law

$$(1 + cj_d)(1 + c'j_d) = (1 + dcc')\left(1 + \frac{c + c'}{1 + dcc'} j_d\right)$$

shows that $\bar{c}$ is additive, and $j_d^2 = d \in F^\times$ provides the order-two factor.

Write ϵ for the nontrivial character of C_2 and ξ for a character of order 3 of C_3. The identity (15) says that R_{E_d} has values 9 at the identity of B_d, 3 at its two nonidentity elements in C_3, and 1 on its three odd elements. Consequently

$$\boxed{\begin{aligned} R_{E_d} &= 1 + \mathrm{reg}_{C_2} + \mathrm{reg}_{C_2 \times C_3} \\ &= 3 \cdot 1 + 2\epsilon + \xi + \xi^2 + \epsilon\xi + \epsilon\xi^2. \end{aligned}} \tag{16}$$

Here the regular representations are inflated from the indicated quotients. All coefficients are nonnegative integers and sum to $3 + 2 + 1 + 1 + 1 + 1 = 9$. This applies to *both* ramified quadratic isomorphism classes, independently of u.

Proposition 6.2 ([**P**] All quadratic torus restrictions pass an integral-character test)**.** *In the wild example $F = \mathbb{Q}_3(\zeta_3)$, $r = 2$, $n = 3$, let D/F be the quaternion division algebra. The function R_E is the character of a nine-dimensional permutation representation of a finite quotient of $E^\times/F^\times$, for every quadratic maximal subfield $E \subset D$. More precisely, for unramified E it is $1 \oplus 2\,\mathbb{C}[C_4]$, and for either ramified E it is $1 \oplus \mathbb{C}[C_2] \oplus \mathbb{C}[C_2 \times C_3]$. For a suitable genuine central character inducing the trivial ordinary central character, the restriction of $L_D\mathbf{1}$ to $\widetilde{E^\times}$ is therefore an* actual genuine representation character *of total dimension* 27*, namely*

E/F	*multiplicities of three-dimensional genuine torus irreducibles*
unramified	$(3, 2, 2, 2)$
ramified, either type	$(3, 2, 1, 1, 1, 1)$.

These multiplicities are invariant under the nontrivial F-automorphism of each E. No irreducibility or even integrality for the full $\widetilde{D^\times}$ is asserted.

Proof. A p-adic field of odd residue characteristic has four square classes; hence F has exactly three quadratic field extensions, one unramified and two ramified. Every quadratic field extension embeds in the quaternion division algebra. Equations (13) and (16) prove the first assertion. Lemma 5.1, together with the degree-three torus Stone–von Neumann theorem, converts those Fourier coefficients into the displayed multiplicities; their weighted dimensions are $3 \cdot 9 = 27$. For the trivial ordinary character one can choose $\widetilde{\theta}$ with $\widetilde{\theta}(a^*) = 1$: the odd-$n$ scalar star kills $\mu_3 \subset F^\times$ (Part I, Section 3), and a faithful character of μ_3 extends across the abelian scalar central group modulo the star image. The nontrivial Galois automorphism acts by inversion on the respective finite quotients' character parameters (with the C_2 factor fixed), so the displayed multiplicities are invariant. □

7 A research plan toward nonabelian gluing

This section records a proposed research program, not further theorems. The immediate target is the following sharply formulated special case.

Target A: the trivial ordinary representation

For $F = \mathbb{Q}_3(\zeta_3)$, D quaternion and $n = 3$, decide whether $L_D\mathbf{1}$ is the character of a single 27-dimensional genuine irreducible of $\widetilde{H}$, with the prescribed central character.

Target B: the full local statement

For every ordinary irreducible ρ with a fixed compatible central character, prove that all coefficients $a_{\rho\tau}$ are integral, or exhibit a specific nonintegral coefficient.

Target C: uniform rank-two theorem

Extend Target B to all odd n, all F of odd residual characteristic with $\mu_n \subset F$, and quaternion D/F.

7.1 Stage 1: make the multiplier and finite level explicit

Use the embedding $D = E_0 \oplus E_0 j \hookrightarrow M_2(E_0)$,

$$x + yj \longmapsto \begin{pmatrix} x & \varpi y \\ \sigma(y) & \sigma(x) \end{pmatrix},$$

to restrict [JC, equation (6.1)] to $D^\times$. Translate the kernel convention $c = 1$ into Part I's relabeled split-side parameter $c_v = 2 \pmod 3$. Check explicitly that the resulting commutator restricts to (10) on both unramified and ramified maximal tori. These comparisons must be carried out at the level of a *cocycle and genuine central character*, not merely at the level of the quotient group $D^\times/F^\times$.

Let $U_D^m = 1 + \mathfrak{P}_D^m$. Seek an open normal congruence subgroup and compatible splitting such that the fixed-central- character class function $L_D\mathbf{1}$ is constant on the appropriate twisted cosets. Prove this factorization at a stated finite level; testing an arbitrary quotient that does not capture $L_D\mathbf{1}$ would be inconclusive. Retain the genuine μ_3 action in the finite central extension or twisted group algebra.

7.2 Stage 2: exact character or Gram calculation

For the resulting finite projective quotient, compute

$$a_{\mathbf{1},\tau} = \langle L_D\mathbf{1}, \Theta_\tau \rangle$$

with exact cyclotomic arithmetic and the normalized fixed-central-character measure. Cross-check its restricted character values against the two tables in Proposition 6.2; these branching rules are necessary constraints, not a uniqueness theorem. A single nonintegral coefficient disproves Target A as stated. If all coefficients at a genuinely capturing level are integers, unitarity and the positive identity value imply Target A. Alternatively compute the twisted positive-semidefinite Gram matrix as a diagnostic; positive semidefiniteness *alone* does not prove Target A.

7.3 Stage 3: construct a candidate genuine representation

Try to induce from a suitable covered compact open subgroup containing a maximal torus and deep principal units. Use the degree-three Stone–von Neumann representation of $\widetilde{E^\times}$ and analyze the action of the normalizer by Clifford theory. A candidate of degree 27 must have restrictions with multiplicities $(3, 2, 2, 2)$ on the unramified torus and $(3, 2, 1, 1, 1, 1)$ on each ramified type. Prove irreducibility by an intertwining criterion and then compare its character on all regular classes with (1). Existence of compatible torus branchings alone does not suffice.

7.4 Stage 4: from one row to all rows

Use the structure of ordinary representations of $D^\times$ (admissible pairs, compact induction, and their character formulas) to transport the construction from $\rho = \mathbf{1}$ to arbitrary ρ. Potential orbit-method arguments on deep p-adic analytic congruence groups require special caution: because $p \mid n$, multiplication by n on the Lie lattice is not an automorphism, so orbit dilation by n^{-1} is

not an established equivalence. A full positivity theorem for all rows would invoke Proposition 3.1 directly.

7.5 Proof checkpoints and stopping criteria

A satisfactory positive solution must supply (i) an explicit cocycle/central-character normalization; (ii) a finite-level factorization theorem or an independent representation construction; (iii) exact full-group character coefficients; and (iv) a uniform all-ρ argument for Target B. A negative answer requires a verifiable nonintegral $a_{\rho\tau}$ at a rigorously justified level. The quadratic toral permutation theorem is already proved and should not be used as a surrogate for either outcome.

8 Interface with the larger Jacquet–Langlands program

The companion problem is local. Even success on Target B would not on its own prove the additional global comparison of ordinary $D^\times$ and the covered group: that comparison remains conditional on matching functions at exceptional split places, complex-place normalization, a global orbital-coefficient identity, and the support/germ hypotheses stated in Part IV. Conversely, the global comparison could be established in a restricted test range without first settling every local irreducible-character row. These projects should be developed in parallel but not conflated.

Acknowledgments and research-status note

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References

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